

## 12.7 Summary and Notes

### 12.7.1 Summary

*Idea.*

1. Confidence sets come from pivots, and are the sets of values that a family of level- $\alpha$  tests accepts (Definition 12.1.1, Proposition 12.1.2).
2. For estimable  $\lambda^\top \beta$ ,  $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} s \sqrt{\lambda^\top G \lambda}$  is exact and free of the generalized inverse (Theorem 12.1.1). A non-estimable function has no confidence set of finite expected length (Proposition 12.1.3). The interval for  $\sigma^2$  is exact under normality and fragile without it (Proposition 12.1.4).
3. The confidence ellipsoid for  $\Lambda^\top \beta$  is the acceptance region of the  $F$  test, shaped by the covariance of the estimates (Theorem 12.2.1); all such ellipsoids are images of one ball around  $\hat{Y}$  (Proposition 12.2.2). Its shadows are Scheffé-type intervals (Lemma 12.2.3, Corollary 12.2.4).
4. The prediction interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{1 + h_0}$  never shrinks below the noise (Theorem 12.3.1). The leverage  $h_0$  is a Mahalanobis distance that reveals hidden extrapolation (Proposition 12.3.2). Without normality prediction intervals stay wrong as  $n \rightarrow \infty$  (Proposition 12.3.3).
5. The Working–Hotelling band, with multiplier  $\sqrt{k F_\alpha(k, n-r)}$ , covers the regression surface over a  $k$ -dimensional family exactly;  $k = 2$  for a straight line (Theorem 12.4.1).
6. Fieller sets for ratios, and calibration sets, are exact, and are intervals, two rays or the whole line (Proposition 12.5.1, Theorem 12.5.3). The classical calibration estimator has no mean (Proposition 12.5.2), and unbounded sets cannot be avoided (Proposition 12.5.4).
7. Conjugate credible sets and predictive intervals have the frequentist shapes (Lemma 12.6.1, Theorem 12.6.2) and equal them under the flat prior (Corollary 12.6.3). Their coverage is nominal on average over the prior, not pointwise (Proposition 12.6.4).

### 12.7.2 Notes and sources

**Confidence intervals and duality.** Confidence intervals, and their construction by inverting tests, are due to Neyman (1937). Lehmann and Romano (2005) develop the duality and its optimality consequences. The two one-sided tests procedure of Exercise (B2) is due to Schuirman (1987). The shortest interval for a normal variance (Exercise (C2)) was worked out by Tate and Klett (1959).

**The source books.** The chapter covers Seber and Lee (2003, chapter 5 and sections 6.1–6.5), Rencher and Schaalje (2008, sections 8.6–8.7) and Sen and Srivastava (1990, section 3.8). These books state the intervals for full-rank models; here they are stated for estimable functions in models of any rank, as in Christensen (2020). Rencher and Schaalje (2008, section 8.7) derive the  $F$  test as a likelihood ratio test, which Chapter 11 covers (Theorem 11.4.1). Hahn and Meeker (1991) is a practical guide to confidence, prediction and tolerance intervals.

**Bands.** The band for a straight line is due to Working and Hotelling (1929), and the general

projection argument to Scheffé (1953). Exact bands over finite intervals are due to Wynn and Bloomfield (1971) and constant-width bands to Gafarian (1964). Miller (1981) surveys the early literature and Liu (2010) gives a modern book-length treatment.

**Normality and prediction.** The warning about conditional coverage after a preliminary test is from Olshen (1973).

**Ratios and calibration.** Fieller's method came from bioassay and is set out in general form in Fieller (1954). Krutchkoff (1967) proposed the inverse estimator and Hoadley (1970) gave its Bayesian interpretation. Osborne (1991) reviews calibration, and Brown (1993) treats the multivariate case. Gleser and Hwang (1987) proved the general form of Proposition 12.5.4.

**Bayesian intervals.** Box and Tiao (1973) and Gelman et al. (2013) treat credible and predictive intervals for the normal linear model under conjugate and flat priors.

**Data.** The murder-rate data are the 2009 US state data of statsmodels; the stack loss data are from Brownlee (1965); the assay is simulated.

### 12.7.3 References

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